Random Access on Narrow Decision Diagrams in External Memory

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SPIN 2024

AARHUS UNIVERSITY

Adiar

I/O-efficient Decision Diagrams

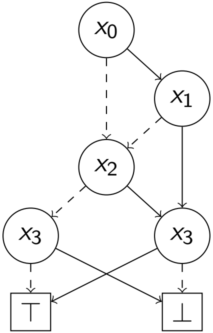
github.com/ssoelvsten/adiar

Features

Optimisations

Features

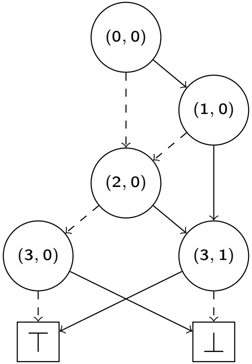
Optimisations



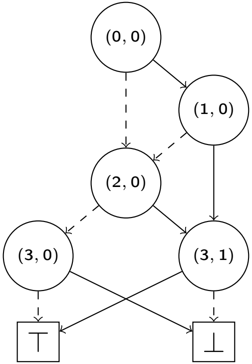
$$
\begin{array}{llr}
[& ((0,0),(2,0),(1,0)) & , \\
 & ((1,0),(2,0),(3,1)) & , \\
 & ((2,0),(3,0),(3,1)) & , \\
 & ((3,0),\top,\bot) & , \\
 & ((3,1),\bot,\top) &]
\end{array}
$$

(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Serialized Representation of a Binary Decision Diagram.



(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

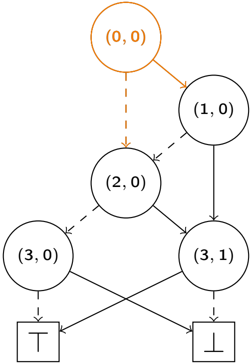


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Priority Queue: Q_{count}:

[

]

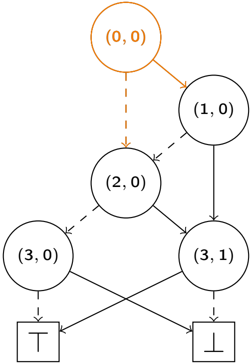


(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

Priority Queue: Q_{count}:

[

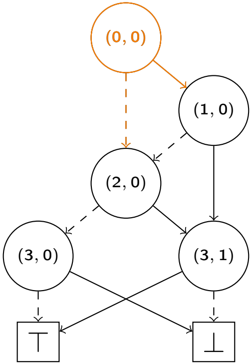
]



(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Priority Queue: Q_{count}:

$$[\quad ((0,0) \xrightarrow{\top} (1,0),\quad 1)\quad ,$$
$$((0,0) \xrightarrow{\perp} (2,0),\quad 1)\quad ,$$
$$]$$

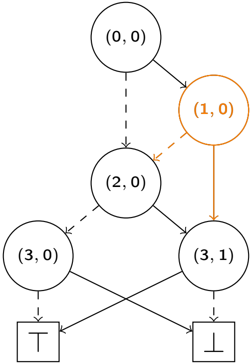


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(1, 0)	0	0

Priority Queue: Q_{count}:

$$[\quad ((0,0) \xrightarrow{\top} (1,0), \quad 1) \quad , \\ ((0,0) \xrightarrow{\perp} (2,0), \quad 1) \quad , \quad]$$

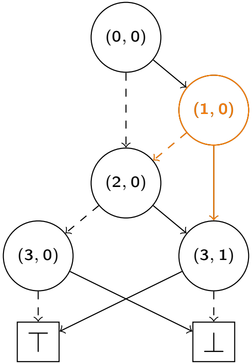


(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

Seek	Sum	Result
(1, 0)	0	0

Priority Queue: Q_{count}:

$$
[\quad ((0,0) \xrightarrow{\top} (1,0),\quad 1)\quad ,
$$
$$
((0,0) \xrightarrow{\bot} (2,0),\quad 1)\quad ,
$$
$$
]
$$

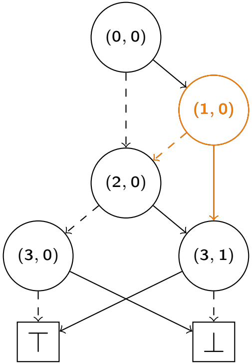


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(1, 0)	1	0

Priority Queue: Q_{count}:

$$[\quad ((0,0) \xrightarrow{\perp} (2,0), \quad 1) \quad , \quad]$$

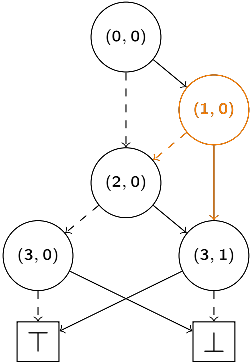


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
$(1, 0)$	1	0

Priority Queue: Q_{count}:

$$
[
\; ((0,0) \xrightarrow{\perp} (2,0),\; 1),\;
((1,0) \xrightarrow{\perp} (2,0),\; 1),\;
((1,0) \xrightarrow{\top} (3,1),\; 1),\;
]
$$

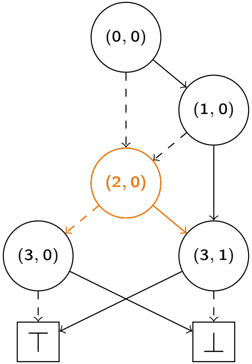


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(2, 0)	0	0

Priority Queue: Q_{count}:

$$[\; ((0,0) \xrightarrow{\perp} (2,0),\; 1),\; ((1,0) \xrightarrow{\perp} (2,0),\; 1),\; ((1,0) \xrightarrow{\top} (3,1),\; 1),\;]$$

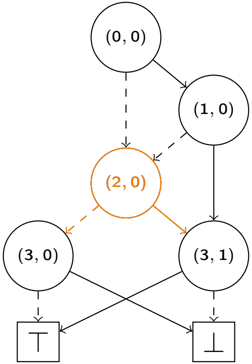


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(2, 0)	0	0

Priority Queue: Q_{count}:

$$
\begin{aligned}
[& \\
&((0,0) \xrightarrow{\perp} (2,0),\quad 1) \quad , \\
&((1,0) \xrightarrow{\perp} (2,0),\quad 1) \quad , \\
& \\
&((1,0) \xrightarrow{\top} (3,1),\quad 1) \quad , \\
&]
\end{aligned}
$$



(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
$(2, 0)$	1	0

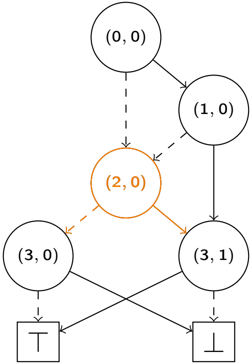
Priority Queue: Q_{count}:

[

$((1,0) \xrightarrow{\perp} (2,0),\quad 1)$,

$((1,0) \xrightarrow{\top} (3,1),\quad 1)$,

]



(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

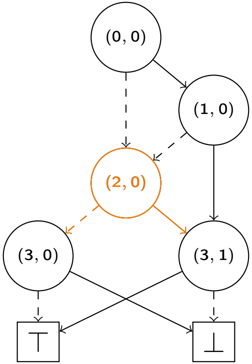
Seek	Sum	Result
(2, 0)	2	0

Priority Queue: Q_{count}:

[

$((1,0) \xrightarrow{\top} (3,1), \quad 1) \quad ,$

]

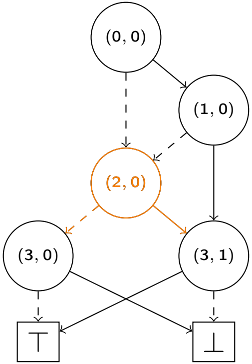


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(2, 0)	2	0

Priority Queue: Q_{count}:

$$[\;((2,0) \xrightarrow{\perp} (3,0),\ 2),\; ((1,0) \xrightarrow{\top} (3,1),\ 1),\; ((2,0) \xrightarrow{\top} (3,1),\ 2)\;]$$

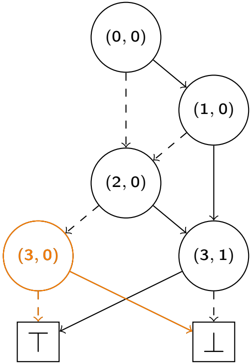


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
$(3,0)$	0	0

Priority Queue: Q_{count}:

$$\begin{aligned}[\;& ((2,0) \xrightarrow{\perp} (3,0), \quad 2) \quad ,\\ & ((1,0) \xrightarrow{\top} (3,1), \quad 1) \quad ,\\ & ((2,0) \xrightarrow{\top} (3,1), \quad 2) \quad]\end{aligned}$$



(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 0)	0	0

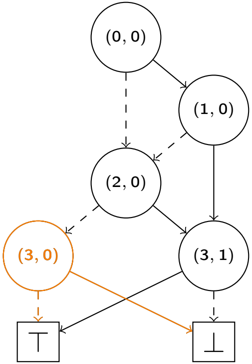
Priority Queue: Q_{count}:

[

$((2,0) \xrightarrow{\perp} (3,0), \quad 2)$,

$((1,0) \xrightarrow{\top} (3,1), \quad 1)$,

$((2,0) \xrightarrow{\top} (3,1), \quad 2)$]

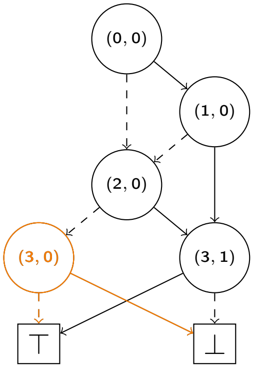


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 0)	2	0

Priority Queue: Q_{count}:

$$[\ ((1,0) \xrightarrow{\top} (3,1),\quad 1)\ ,$$
$$((2,0) \xrightarrow{\top} (3,1),\quad 2)\ ]$$



(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

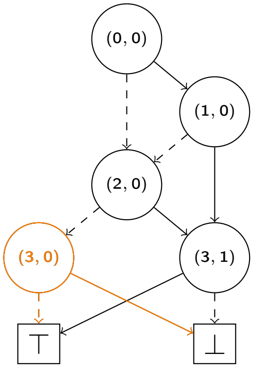
Seek	Sum	Result
(3, 0)	2	2

Priority Queue: Q_{count}:

[

$((1, 0) \xrightarrow{\top} (3, 1), \quad 1)$,

$((2, 0) \xrightarrow{\top} (3, 1), \quad 2)$]

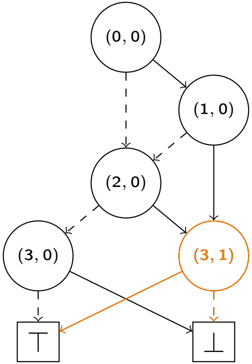


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 1)	0	2

Priority Queue: Q_{count}:

$$[\ ((1,0) \xrightarrow{\top} (3,1),\quad 1),\quad ((2,0) \xrightarrow{\top} (3,1),\quad 2)\ ]$$

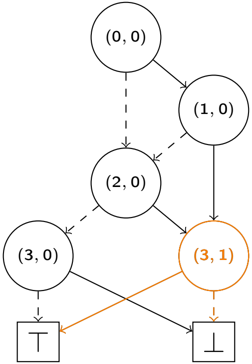


(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 1)	0	2

Priority Queue: Q_{count}:

$$[\ ((1,0) \xrightarrow{\top} (3,1),\quad 1)\ ,\ ((2,0) \xrightarrow{\top} (3,1),\quad 2)\ ]$$



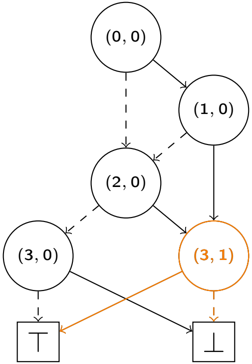
(a) $(x_0 \wedge x_1 \wedge x_3) \vee (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 1)	1	2

Priority Queue: Q_{count}:

[

$((2,0) \xrightarrow{\top} (3,1), \quad 2)$]



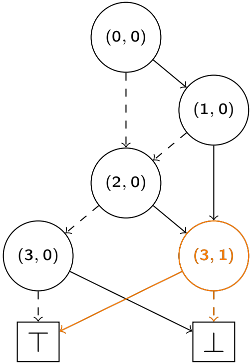
(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 1)	3	2

Priority Queue: Q_{count}:

[

]

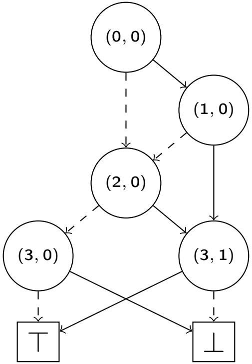


(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

Seek	Sum	Result
(3, 1)	3	5

Priority Queue: Q_{count}:

[ ]



(a) $(x_0 \land x_1 \land x_3) \lor (x_2 \oplus x_3)$

Result

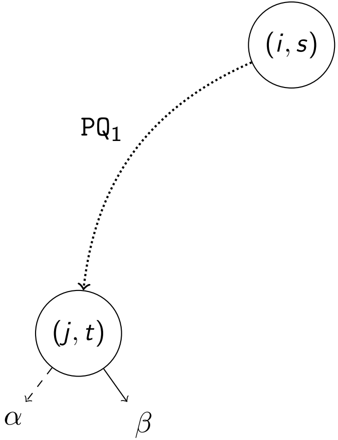
5

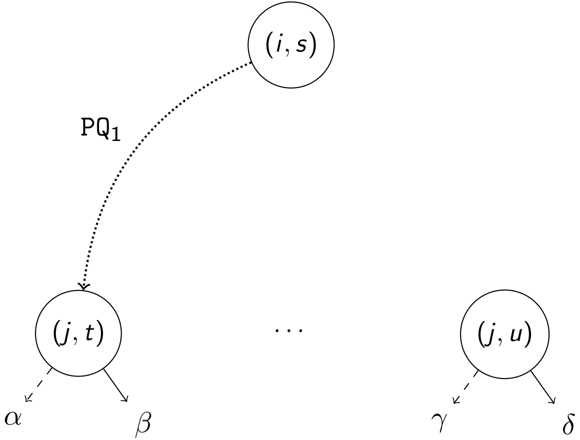
Priority Queue: Q_{count}:

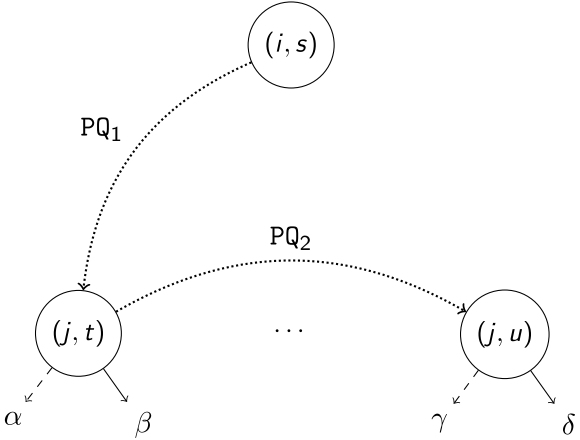
[

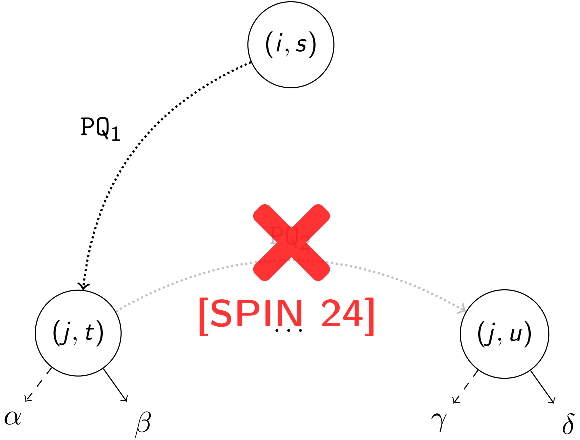
]

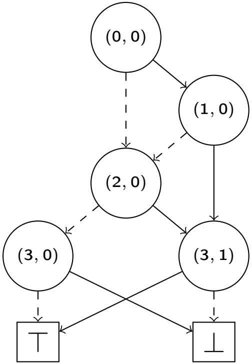
(i, s)

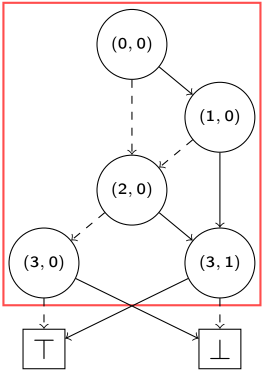

(i, s)
$\mathtt{PQ}_1$
(j, t)
α
β

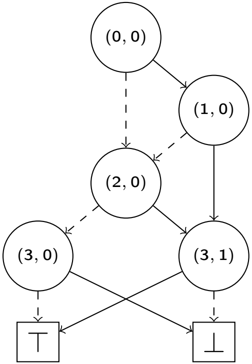

(i, s)
$\mathtt{PQ}_1$
(j, t)
$\cdots$
(j, u)
α
β
γ
δ

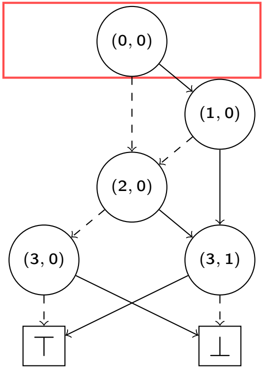

(i, s)
$\mathtt{PQ}_1$
$\mathtt{PQ}_2$
(j, t)
$\cdots$
(j, u)
α
β
γ
δ

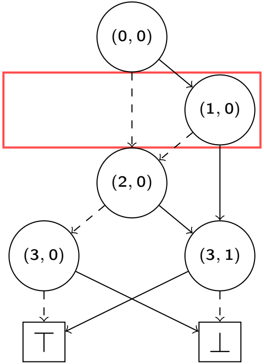

(i, s)
$\mathtt{PQ}_1$
$\mathtt{PQ}_2$
[SPIN 24]
(j, t)
$\dots$
(j, u)
α
β
γ
δ

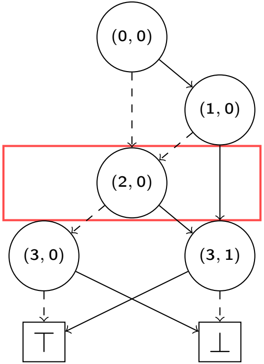

(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥

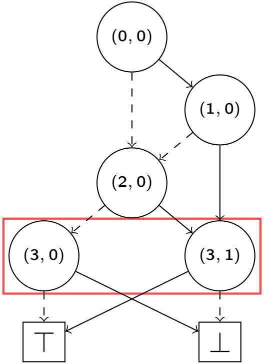

(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥

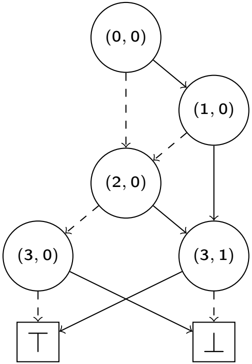

(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥

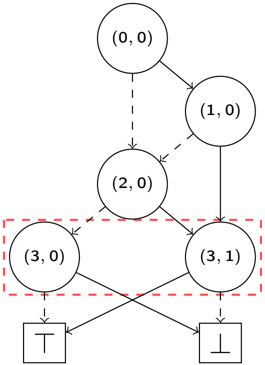

(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥


(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥

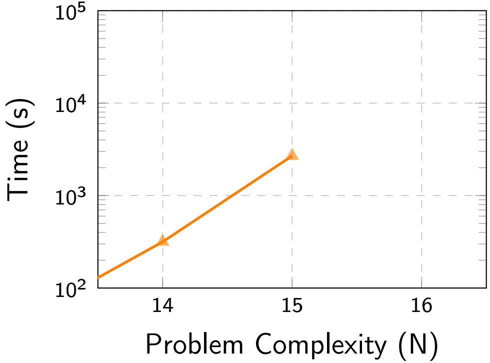

(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥


(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥


(0, 0)
(1, 0)
(2, 0)
(3, 0)
(3, 1)
⊤
⊥

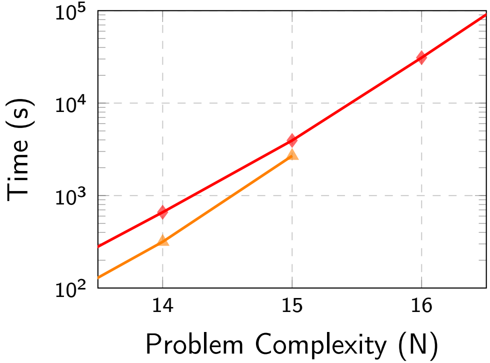


$$\frac{\text{Width}}{2}$$



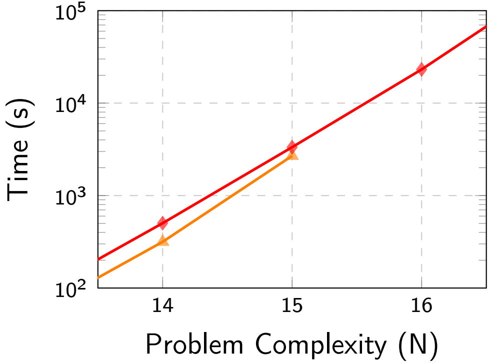
			$N = 15$
△	CUDD	v3.0	: 44.8 min

Queens | 300 GiB of RAM



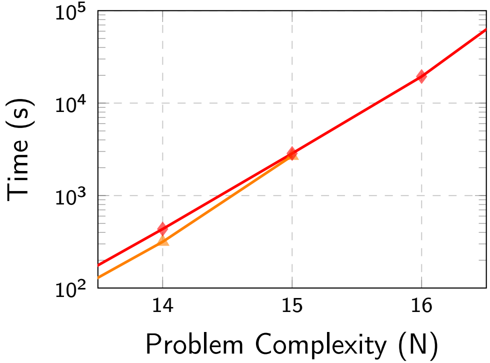
				$N = 15$
△	CUDD	v3.0	:	44.8 min
◇	Adiar	v1.0	:	66.7 min

Queens | 300 GiB of RAM



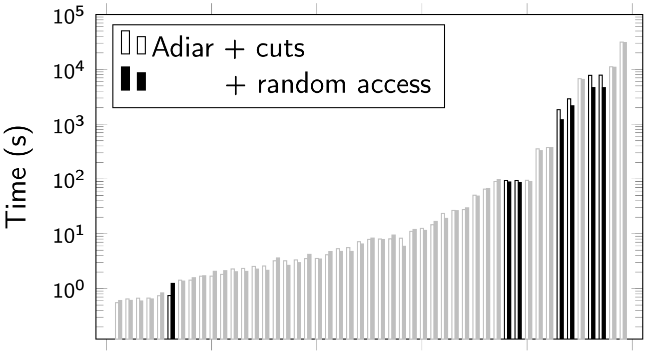
				$N = 15$
△	CUDD	v3.0	:	44.8 min
◇	Adiar	v1.0	:	66.7 min
	+ cuts		:	56.8 min

Queens | 300 GiB of RAM



				$N = 15$
△	CUDD	v3.0	:	44.8 min
◇	Adiar	v1.0	:	66.7 min
	+ cuts		:	56.8 min
	+ random access		:	47.2 min

Queens | 300 GiB of RAM



Speed Ups

6 (+26.9%)

Slowdowns

1 (-66.8%)

EPFL Circuit Verification | 300 GiB of RAM

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ssoelvsten.github.io

Adiar

github.com/ssoelvsten/adiar

ssoelvsten.github.io/adiar

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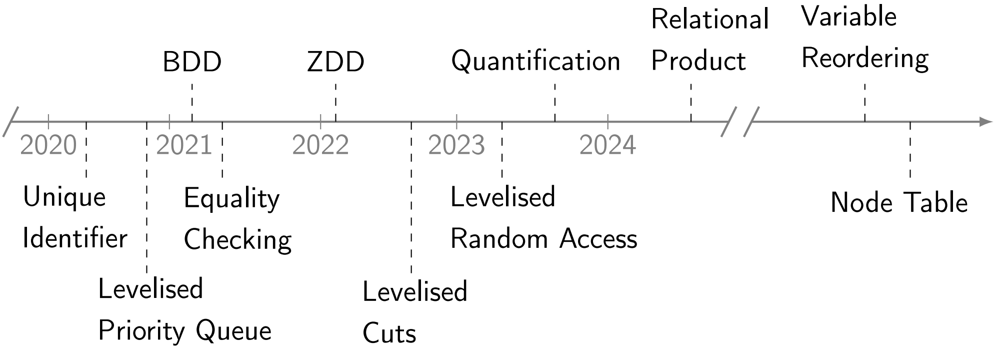
Relational Product
Variable Reordering
BDD
ZDD
Quantification
2020
2021
2022
2023
2024
Unique Identifier
Equality Checking
Levelised Random Access
Node Table
Levelised Priority Queue
Levelised Cuts